

Software Requirements Specification (SRS)

Event Management System

Project: Event Management System

Description: A system to manage and organize events with registrations.

Version: 1.0

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Revision History

Version	Date	Author	Change summary	Approval
1.0	03-09-2026	Team 7	Initial SRS drafted for Event Management System	Pending

Approvals

Role	Name	Signature / Email	Date

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Event Management System (EMS), a web application for creating, publishing, and managing events, and for handling attendee registration, payment, and ticketing. It defines the functional and non-functional requirements, external interfaces, and verification criteria intended for the development team, QA engineers, and stakeholders.

1.2 Scope

The EMS enables Event Organizers to create and publish events with configurable ticket types and capacities, enables Attendees to discover events, register, and pay for tickets, and enables Administrators to manage users and monitor the platform. The system is implemented as a web application using Python (Django) for the backend, a relational database, and integration with a third-party payment gateway and email service. The scope excludes on-site hardware such as physical badge printers or turnstile access control, which may be addressed in a future phase.

1.3 Audience

Backend and frontend developers, QA engineers, DevOps/system administrators, project stakeholders (event organizers, business owners), and security/compliance reviewers.

1.4 Definitions

List of acronyms: EMS (Event Management System), UI (User Interface), API (Application Programming Interface), RBAC (Role-Based Access Control), PII (Personally Identifiable Information), PCI-DSS (Payment Card Industry Data Security Standard), TLS (Transport Layer Security), QR (Quick Response code), CSV (Comma-Separated Values), ORM (Object-Relational Mapping).

2. Overall description

2.1 Product perspective

The EMS is a standalone, self-contained web application built on the Django framework, following a client-server architecture. It exposes a browser-based UI (and optionally a REST API for future mobile clients), a Django application server, a relational database (PostgreSQL), a background task queue (Celery/Redis) for asynchronous notifications, and integrations with a payment gateway and transactional email provider.

2.2 Major product functions

- User registration, email verification, and authentication
- Role-based access for Attendee, Organizer, and Administrator
- Event creation, editing, publishing, and capacity/ticket-type management
- Event discovery via browse and search
- Event registration and online payment
- Automated e-ticket generation with QR codes
- Automated email confirmations and reminders
- Reporting and CSV export for organizers
- User account administration and system monitoring

2.3 User roles and characteristics

- Attendee: general public user with basic web/mobile literacy; wants a fast, simple registration and payment flow.

- Event Organizer: creates and manages events; needs a straightforward dashboard for event setup, attendee tracking, and reporting.
- System Administrator: manages platform users, security configuration, and operational health; needs monitoring and audit tools.

2.4 Operating environment

Server-side: Linux-based cloud hosting running Python 3.x / Django, PostgreSQL database, Redis for caching and task queuing, deployed behind a load balancer with HTTPS termination. Client-side: modern desktop and mobile web browsers (Chrome, Firefox, Safari, Edge, current and previous major versions).

2.5 Constraints

PCI-DSS considerations apply to payment handling (delegated to a certified third-party gateway); applicable data-protection regulations (e.g., GDPR-equivalent principles) govern storage of attendee PII; the system must be built using Python/Django per the specified technology stack; email deliverability depends on a third-party transactional email provider.

3. External interface requirements

3.1 User interfaces

A responsive, browser-based UI supporting desktop and mobile breakpoints. Public pages (event browse/search/detail) require no login; registration, payment, and dashboards require authentication. Accessibility target: WCAG 2.1 AA where applicable, including keyboard navigation and screen-reader-friendly labels.

3.2 Hardware interfaces

No dedicated hardware is required. The system is designed to run on standard cloud server infrastructure and to be accessed from standard end-user devices (desktop/laptop computers, tablets, and smartphones) with no specialised peripherals.

3.3 Software interfaces

- Payment Gateway API (e.g., Stripe/Razorpay) over HTTPS for authorization, capture, and refunds.
- Transactional Email API (e.g., SendGrid/SES) for confirmations and reminders.
- Optional Calendar export (iCal) for attendees to add events to personal calendars.
- Django REST Framework API for potential future mobile or third-party integrations.

3.4 Communications

- TLS 1.2+ enforced for all client-server and server-to-gateway communication.
- Webhook callbacks from the payment gateway are verified via signed payloads before processing.
- Retry/backoff strategy for outbound notification and payment API calls.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow EMS-F-####. This document defines 22 functional requirements, 5 non-functional requirements, 3 security objectives, and 5 security requirements.

4.1 User Registration & Authentication

Description: Allow users to create accounts, verify their identity, log in securely, and operate under role-based permissions.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-001	The system shall allow a new user to register an account using name, email address, and password.	Functional	High	All users	AC-EMS-F-001: Valid registration form creates an account and sends a verification email. Test: TC-Auth-01	Django auth / allauth
EMS-F-002	The system shall verify a user's email address via a time-limited confirmation link before allowing login.	Functional	High	Security	AC-EMS-F-002: Unverified accounts cannot log in; link expires after 24h. Test: TC-Auth-02	Depends on Email service
EMS-F-003	The system shall authenticate registered users via email and password and issue a session token.	Functional	High	All users	AC-EMS-F-003: Correct credentials grant access; incorrect credentials are rejected with a generic error. Test: TC-Auth-03	Django session/JWT
EMS-F-004	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and record an audit event.	Functional	High	Security	AC-EMS-F-004: 6th attempt within lockout window is rejected with lockout message. Test: TC-Auth-04	Rate limiting middleware
EMS-F-005	The system shall support role-based access control distinguishing Attendee, Organizer, and Administrator roles.	Functional	High	System Architect	AC-EMS-F-005: Users only see menu items and API endpoints permitted for their role. Test: TC-Auth-05	Django groups/permissions

4.2 Event Management (Organizer)

Description: Allow Organizers to create, configure, and publish events, including ticket types and capacity limits.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-010	The system shall allow an Organizer to create a new event with title, description, category, venue, date/time, and capacity.	Functional	High	Event Organizer	AC-EMS-F-010: Event saved with status 'Draft' and appears in organizer dashboard. Test: TC-Evt-01	Django ORM model
EMS-F-011	The system shall allow an Organizer to edit event details prior to the event start date.	Functional	High	Event Organizer	AC-EMS-F-011: Changes persist and are reflected on the public event page within 1 minute. Test: TC-Evt-02	
EMS-F-012	The system shall allow an Organizer to publish or unpublish an event, controlling its visibility to attendees.	Functional	High	Event Organizer	AC-EMS-F-012: Unpublished events do not appear in search/browse results. Test: TC-Evt-03	
EMS-F-013	The system shall allow an Organizer to define one or more ticket types per event, each with a	Functional	High	Event Organizer	AC-EMS-F-013: Ticket types are stored and offered during registration; sales	

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	name, price, and quantity limit.				stop when quantity is exhausted. Test: TC-Evt-04	
EMS-F-014	The system shall prevent event registrations from exceeding the configured event or ticket-type capacity.	Functional	High	Business	AC-EMS-F-014: Registration attempt beyond capacity is rejected and 'Sold Out' is displayed. Test: TC-Evt-05	Requires row-level locking to avoid overselling

4.3 Event Discovery & Registration (Attendee)

Description: Allow attendees to find events and register for them, and manage their own registrations.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-020	The system shall allow any visitor to browse and search published events by keyword, category, date, and location.	Functional	High	Attendee	AC-EMS-F-020: Search returns matching published events within 2 seconds. Test: TC-Disc-01	
EMS-F-021	The system shall display full event details, including agenda, venue map link, ticket types, and remaining availability.	Functional	Medium	Attendee	AC-EMS-F-021: Event detail page renders all populated fields correctly. Test: TC-Disc-02	
EMS-F-022	The system shall allow an authenticated Attendee to register for an event by selecting a ticket type and quantity.	Functional	High	Attendee	AC-EMS-F-022: Successful registration creates a registration record in 'Pending Payment' or 'Confirmed' status. Test: TC-Reg-01	Includes EMS-F-030
EMS-F-023	The system shall allow an Attendee to cancel a registration up to a configurable cutoff time before the event.	Functional	Medium	Attendee	AC-EMS-F-023: Cancelled registration frees capacity and updates status to 'Cancelled'. Test: TC-Reg-02	
EMS-F-024	The system shall allow an Attendee to view a list of their current and past registrations from a personal dashboard.	Functional	Medium	Attendee	AC-EMS-F-024: Dashboard lists all registrations with status and event date. Test: TC-Reg-03	

4.4 Payment & Ticketing

Description: Handle secure payment processing, e-ticket issuance, and refunds.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-030	The system shall process ticket payments through an integrated payment gateway supporting major credit/debit cards.	Functional	High	Business / Finance	AC-EMS-F-030: Successful payment transitions registration to 'Confirmed'; failed	Stripe/Razorpay gateway

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
					payment leaves it 'Pending Payment'. Test: TC-Pay-01	
EMS-F-031	The system shall generate a unique QR-coded e-ticket and email it to the attendee upon confirmed registration.	Functional	High	Attendee / Operations	AC-EMS-F-031: E-ticket PDF with valid QR code is attached to the confirmation email. Test: TC-Pay-02	Depends on EMS-F-040
EMS-F-032	The system shall support full or partial refunds for cancelled registrations according to the event's refund policy.	Functional	Medium	Business / Finance	AC-EMS-F-032: Refund amount matches policy rules and is reflected in the payment gateway and registration record. Test: TC-Pay-03	

4.5 Notifications & Reminders

Description: Keep attendees informed via automated, timed communications.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-040	The system shall send an automated email confirmation immediately after successful event registration.	Functional	High	Attendee	AC-EMS-F-040: Confirmation email delivered within 2 minutes of registration. Test: TC-Notif-01	
EMS-F-041	The system shall send automated reminder notifications 48 hours and 2 hours before an event to all confirmed attendees.	Functional	Medium	Attendee / Operations	AC-EMS-F-041: Reminder emails are queued and sent at the configured offsets. Test: TC-Notif-02	Scheduled via Celery beat

4.6 Reporting & Administration

Description: Provide organizers and administrators with reporting and account-management capabilities.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-050	The system shall allow an Organizer to generate an attendee list and revenue report for a given event, exportable as CSV.	Functional	Medium	Event Organizer	AC-EMS-F-050: Exported CSV matches on-screen attendee and revenue data. Test: TC-Rep-01	
EMS-F-051	The system shall allow an Administrator to view, activate, deactivate, and change the role of any user account.	Functional	Medium	System Administrator	AC-EMS-F-051: Deactivated accounts cannot log in; role changes take effect immediately. Test: TC-Adm-01	

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow EMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
EMS-NF-001	Event search and listing pages shall render in ≤ 2 seconds (95th percentile) under normal load (≤ 500 concurrent users).	Performance	High	Load test report shows 95th percentile ≤ 2 s. Test: TC-Perf-01
EMS-NF-002	The system shall be available 99.5% of the time monthly, excluding scheduled maintenance windows.	Reliability/Availability	High	Uptime monitoring reports $\geq 99.5\%$ per month. Test: Ops reports
EMS-NF-003	The system shall support at least 500 concurrent registration attempts during peak usage without data loss, duplicate registrations, or overselling.	Scalability	High	Load test with 10,000 concurrent registration attempts shows zero overselling. Test: TC-Perf-02
EMS-NF-004	The user interface shall be responsive and usable on desktop, tablet, and mobile screen widths from 360px upward.	Usability	Medium	UI verified on reference breakpoints (360px, 768px, 1440px). Test: TC-UX-01
EMS-NF-005	The system shall retain event, registration, and payment audit logs for a minimum of 3 years for financial and compliance purposes.	Audit/Data Retention	Medium	Archived logs retrievable after 3-year simulated retention. Test: TC-OPS-01

5.1 Security

5.1.1 Security Objectives

- Protect attendee personal data (PII) and payment information from unauthorised access, disclosure, or tampering, in line with applicable data-protection regulations.
- Ensure only authenticated, appropriately-authorised users (Attendee, Organizer, Administrator) can perform actions within their assigned role.
- Maintain the integrity and availability of event, registration, and payment data against accidental or malicious modification.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
EMS-SR-001	All network communication between client, application server, and payment gateway shall use TLS 1.2 or higher.	Security	High	AC-EMS-SR-001: SSL scan shows no protocol below TLS 1.2 accepted. Test: TC-Sec-01
EMS-SR-002	The system shall never store raw credit card numbers; payment card data shall be tokenised by the PCI-DSS-compliant payment gateway.	Security	High	AC-EMS-SR-002: Database inspection confirms no PAN fields; only gateway tokens are stored. Test: TC-Sec-02
EMS-SR-003	The system shall hash and salt user passwords using a modern algorithm (e.g., Argon2 or PBKDF2) and never store passwords in plaintext.	Security	High	AC-EMS-SR-003: Password column inspection confirms salted hashes only. Test: TC-Sec-03
EMS-SR-004	The system shall enforce role-based access control on every API endpoint, rejecting requests from users without the required role.	Security	High	AC-EMS-SR-004: Automated authorization test suite confirms 403 responses for out-of-role requests. Test: TC-Sec-04

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
EMS-SR-005	The system shall log and alert on suspicious activity, including repeated failed logins and abnormal registration rates, for review by an Administrator.	Security	Medium	AC-EMS-SR-005: Simulated attack pattern generates an alert visible in the admin monitoring panel. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

Exit criteria for acceptance:

All high-priority functional requirements are implemented and verified, no critical NFR or security-requirement failures remain open, and the RTM shows all associated test cases passed.

Acceptance test suites:

- Authentication & Authorization
- Event Management (Create/Edit/Publish)
- Event Discovery & Registration
- Payment & Refunds
- Notifications
- Reporting
- Performance & Scalability
- Security
- Accessibility

7. System models and diagrams

7.1 UML Use-Case Diagrams

The diagrams below illustrate the primary actors and their interactions with the Event Management System.

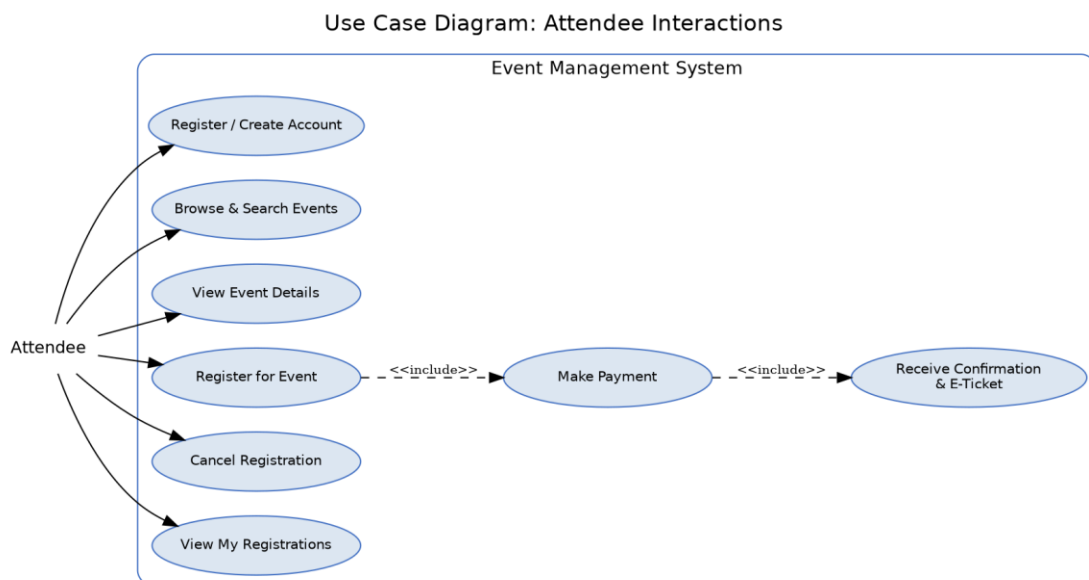


Figure 1: Use-case diagram for Attendee interactions.

Use Case Diagram: Organizer & Administrator Interactions

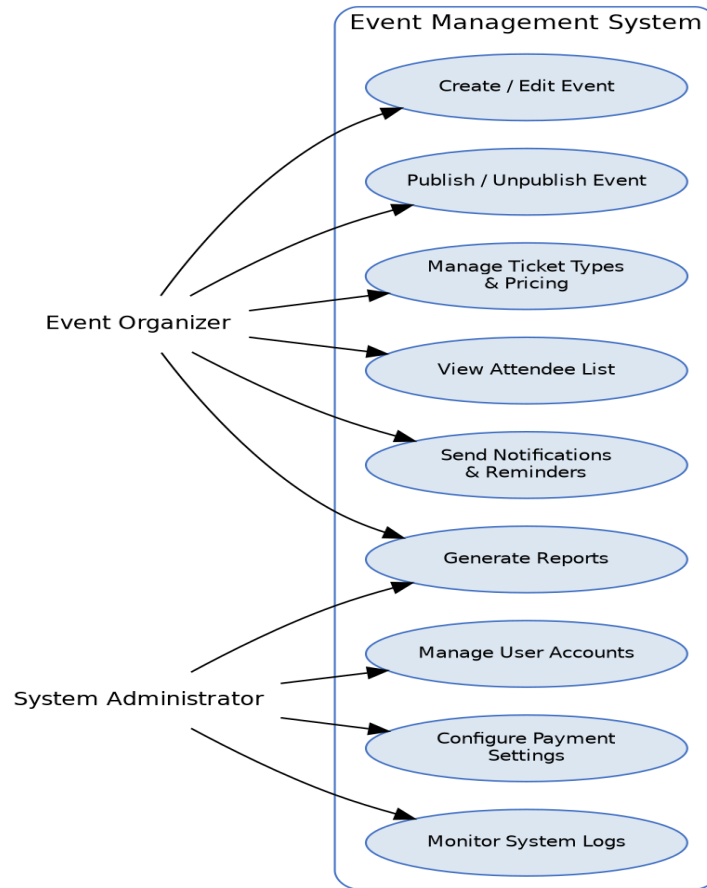


Figure 2: Use-case diagram for Event Organizer and System Administrator interactions.

8. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix (RTM) traces all functional, non-functional, and security requirements to their corresponding design specifications, implementation modules, and test cases. The RTM will be updated throughout implementation and testing to maintain complete requirements coverage.

Req ID	Requirement (Short)	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
EMS-F-001	User registration	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	—
EMS-F-002	Email verification	4.1 / DS-Auth-01	AuthModule / Email	TC-Auth-02	N	Depends on Email service
EMS-F-003	User login	4.1 / DS-Auth-01	AuthModule	TC-Auth-03	N	Django session/JWT
EMS-F-004	Account lockout	4.1 / DS-Auth-02	AuthModule	TC-Auth-04	N	Rate limiting middleware
EMS-F-005	Role-based access	4.1 / DS-Auth-02	AuthModule	TC-Auth-05	N	Django groups/permissions
EMS-F-010	Create event	4.2 / DS-Event-01	EventModule	TC-Evt-01	N	Django ORM model

EMS-F-011	Edit event	4.2 / DS-Event-01	EventModule	TC-Evt-02	N	—
EMS-F-012	Publish/unpublish event	4.2 / DS-Event-01	EventModule	TC-Evt-03	N	Controls event visibility
EMS-F-013	Define ticket types	4.2 / DS-Event-02	EventModule	TC-Evt-04	N	Name, price and quantity
EMS-F-014	Enforce event capacity	4.2 / DS-Event-03	EventModule / DB	TC-Evt-05	N	Row-level locking required
EMS-F-020	Browse/search events	4.3 / DS-Disc-01	CatalogModule	TC-Disc-01	N	Search by keyword/category/date/location
EMS-F-021	Display event details	4.3 / DS-Disc-01	CatalogModule	TC-Disc-02	N	Includes venue, agenda and availability
EMS-F-022	Register for event	4.3 / DS-Reg-01	RegistrationModule	TC-Reg-01	N	Includes payment workflow
EMS-F-023	Cancel registration	4.3 / DS-Reg-01	RegistrationModule	TC-Reg-02	N	Configurable cancellation cutoff
EMS-F-024	View registrations	4.3 / DS-Reg-01	RegistrationModule	TC-Reg-03	N	Current and past registrations
EMS-F-030	Process payment	4.4 / DS-Pay-01	PaymentModule	TC-Pay-01	N	Third-party gateway
EMS-F-031	Generate e-ticket	4.4 / DS-Pay-02	TicketingModule	TC-Pay-02	N	QR-coded PDF
EMS-F-032	Process refunds	4.4 / DS-Pay-01	PaymentModule	TC-Pay-03	N	Based on refund policy
EMS-F-040	Registration confirmation email	4.5 / DS-Notif-01	NotificationModule	TC-Notif-01	N	Sent after successful registration
EMS-F-041	Event reminder notifications	4.5 / DS-Notif-01	NotificationModule	TC-Notif-02	N	48h and 2h before event
EMS-F-050	Attendee/revenue report	4.6 / DS-Rep-01	ReportingModule	TC-Rep-01	N	CSV export

8. Requirements Traceability Matrix (RTM) - Continued

The following entries complete the RTM by adding the remaining functional requirement and all non-functional and security requirements.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status	Comments
EMS-F-051	Manage user accounts	4.6 / DS-Adm-01	AdminModule	TC-Adm-01	N	Activate/deactivate users and change roles
EMS-NF-001	Search response time	5 / DS-Perf-01	CatalogModule / Web	TC-Perf-01	N	95th percentile response time <= 2 seconds
EMS-NF-002	Monthly availability	5 / DS-Rel-01	Infrastructure / Web	Ops reports	N	Target availability: 99.5%
EMS-NF-003	Peak registration scalability	5 / DS-Perf-02	RegistrationModule / DB	TC-Perf-02	N	No data loss, duplicates, or overselling
EMS-NF-004	Responsive user interface	5 / DS-UX-01	WebUI	TC-UX-01	N	Test at 360px, 768px, and 1440px
EMS-NF-005	Audit-log retention	5 / DS-Ops-01	AuditModule / Storage	TC-OPS-01	N	Minimum retention period: 3 years
EMS-SR-001	TLS 1.2 or higher	5.1.2 / DS-Sec-01	Infrastructure / Web	TC-Sec-01	N	Reject protocols below TLS 1.2
EMS-SR-002	No raw card-number storage	5.1.2 / DS-Sec-02	PaymentModule	TC-Sec-02	N	Store gateway tokens only
EMS-SR-003	Secure password hashing	5.1.2 / DS-Sec-03	AuthModule	TC-Sec-03	N	Salted Argon2 or PBKDF2 hashes
EMS-SR-004	RBAC on every API endpoint	5.1.2 / DS-Sec-04	AuthModule / API	TC-Sec-04	N	Unauthorized role receives HTTP 403
EMS-SR-005	Suspicious-activity alerts	5.1.2 / DS-Sec-05	MonitoringModule	TC-Sec-05	N	Alert visible in administrator panel